

Outline of Comments: FDA Draft Guidance - *Homologous Use of Human Cells, Tissues, and Cellular and Tissue-Based Products*

FDA Guidance Provision	Comment	Rationale
FDA's definition of "homologous" for purposes of applying either 361 (1271) or 351 (full BLA) is: the repair, reconstruction, replacement, or supplementation of a recipient's cells or tissues with an HCT/P that performs the same basic function or functions in the recipient as in the donor (21 CFR §1271.3(c)) HCT/P (as regulated solely by 361) is intended for homologous use only, as reflected by the labeling, advertising, or other indications of the manufacturer's objective intent (§1271.10(a)(2))	This regulatory standard is broad enough to incorporate the fundamental properties of stem cells, e.g. their ability to self-renew, as well as proliferate and differentiate into lineage-committed functional cells to help achieve or restore the same or similar functions in the recipient as in the donor. Instead, FDA opted to interpret this flexible standard so narrowly that the use of most, if not all, differentiating stem cells (e.g., mesenchymal from stroma or adipose tissue, hematopoietic, embryonic or induced pluripotent) is deemed non-homologous Standard should not be the cell function in the tissue or location in which it is identified, but the function or characteristics of the cells themselves. They should be deemed homologous under this standard if they help achieve or restore the same or similar function in the recipient as in the donor	This broader interpretation would permit the safe and documented restorative benefits of stem cells with all the protections that exist under the 1271 regulations.

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	FDA should also view the European Medicines Agency reinterpretation by viewing not the original function of these cells, but whether they are used in “the same anatomical or histological environment”	
Stated examples of homologous use include recipient cells that are identical to donor cells (e.g., skin for skin); or non-identical cells that perform one or more of the same basic functions in recipient as in donor Heart valve replacing heart valve; or Pericardium to repair or reconstruct dura matter and serve as wound covering (Guidance at 3-4)	The stated examples are the most narrow and most obvious and also typically utilized in a “same surgical procedure” situation. FDA alone knows the specific facts and circumstances of HCT/P uses that it has determined to be homologous. Broader and more far reaching examples should be provided, not merely the narrowest and most obvious examples. For example we are aware of a homologous determination made for the use of amniotic membrane tissue for use in healing corneal erosion (see Protera-BioTissue)	
Not homologous if HCT/Ps “for use as an unproven treatment for a myriad of disease conditions.” (Guidance at 4)	This states the obvious and is not helpful.	
Examples cited for homologous “repair, reconstruction, replacement, or	The Guidance should be broadened to identify permitted use of differentiating	

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supplementation of recipient's cells or tissues" include implantation of dermal matrix into facial wrinkles, or use of bone chips to supplement bony defects.	stem cells for which ample research exists to demonstrate use to restore the same of similar functions in the recipient as in the donor.	
Likewise, FDA states that "[i]t is not necessary for the HCT/P in the recipient to perform all of the basic functions it performed in the donor" to meet the definition of homologous use, but any basic function that the HCT/P is expected to perform in the recipient must be a basic function that the HCT/P performed in the donor. (pg. 4-5). The example given for homologous use was the use of hematopoietic stem/progenitor cells (HPC) for disorders affecting the hematopoietic system because the peripheral blood product performs the same basic function of reconstituting the hematopoietic system in the recipient. Cord blood HPCs infused intravenously to differentiate into neuronal cells for treatment of cerebral palsy were given as an example of non-homologous use because of "insufficient evidence to support that such differentiation is a basic function of these cells in the donor."	FDA is again using unreasonably narrow examples to deny 1271 treatment for technologies that have been proven safe and effective. For example, clinicians generally agree that the mechanism of action for the use of HPCs for blood reconstitution is well known and well characterized. This mechanism of action involves engraftment, massive cell proliferation and differentiation. That scientific evidence includes: <i>in vitro</i> cell cultures demonstrating that progenitor cells can grow and differentiate; animal models following lethal radiation demonstrated rescue with bone marrow or even highly purified cells given by IV where blood cell counts eventually recover to normal levels. Human studies using genetic marking show that new blood cells are derived from donor stem cells. Stem cell studies for cardiac repair have demonstrated direct differentiation, fusion, reduced inflammation, stimulation of endogenous repair and	

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	improved blood flow. It is likely that cell therapy produces a combination of these effects based on the specific clinical situation. One of the advantages of cell therapy over drugs is the ‘intelligence’ of the cell and it’s ability to assess, adapt to and respond to specific signaling in damaged tissue. In the case of using cord blood cells for CP, there is no expectation of neuronal differentiation or integration. Rather, the monocytes in cord blood have been documented to act through paracrine signaling to reduce inflammation and promote neural connectivity. Finally, dozens of human clinical studies have documented the use of regulatory T-cells as adaptive cell therapy for the prevention of graft v. host disease (GVHD) in human transplant patients. If these cells or tissues possess the characteristics to proliferate and differentiate, FDA has the discretion to opine that these characteristics allow them to perform these functions in the donor or the recipient.	

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The basic function of a corneal graft transplanted to restore sight is a homologous use, but amniotic membrane used for bone tissue replacement to support bone regeneration is not a homologous use.	In addition to the example of a non-homologous use of amniotic membrane, the Guidance should be modified to include a homologous use of this tissue (e.g., use in healing corneal erosion (<i>see</i> Protera-BioTissue))	
A HCT/P may perform the same basic function even when it is not used in the same anatomic location where it existed as a donor. Use of an acellular dermal product for supplemental support, protection, reinforcement or covering for a tendon was given as an example of a homologous use, while use for tendon replacement or repair was non homologous. Also use of amniotic membrane for wound healing of dermal ulcers was non-homologous.	In addition to the example of a non-homologous use of amniotic membrane, for example, the Guidance should be modified to include a homologous use of this tissue (e.g., use in healing corneal erosion (<i>see</i> Protera-BioTissue)) Furthermore, if FDA could modify its position to consider the function of the cell or tissue, regardless of location, use of amniotic membrane for wound healing should be considered homologous.	